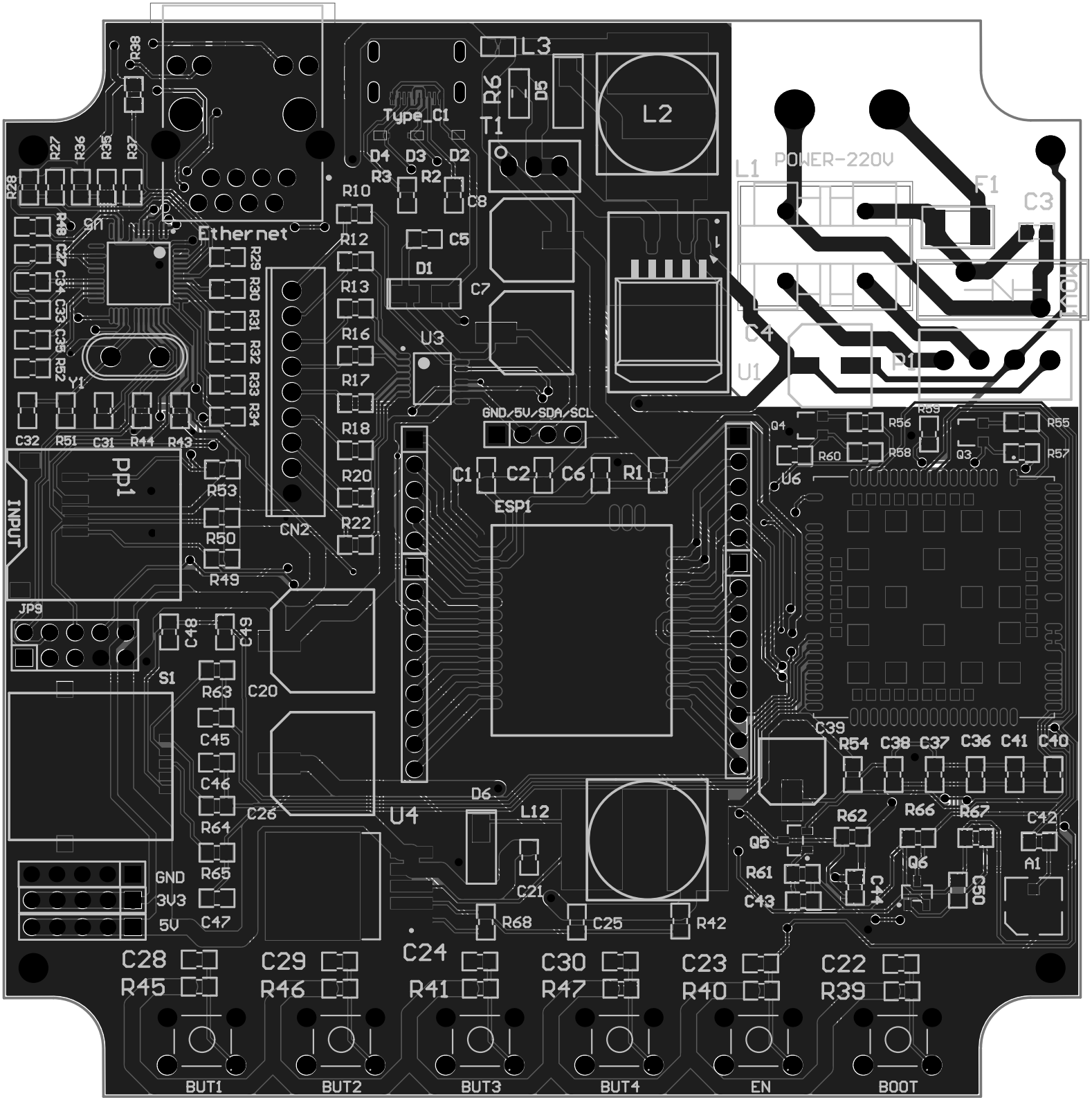

L3
L2
R6
D5
T1
Type_C1
D4
D3
D2
R3
R2
C8
C5
D1
C7
U3
R10
R12
R13
R16
R17
R18
R20
R22
CN2
Ethernet
SH
R38
R27
R36
R35
R37
R28
R48
C27
C34
C33
C35
R52
R29
R30
R31
R32
R33
R34
Y1
C32
R51
C31
R44
R43
pd1
INPUT
R53
R50
R49
JP9
C48
C49
S1
R63
C20
C45
C46
R64
C26
R65
C47
GND
3V3
5V
U4
D6
L12
C21
R68
C25
R42
C24
C28
R45
C29
R46
R41
C30
R47
C23
R40
C22
R39
BUT1
BUT2
BUT3
BUT4
EN
BOOT
GND/5V/SDA/SCL
C1
C2
C6
R1
ESP1
L1
POWER-220V
F1
C3
C4
U1
P1
Q4
R56
R59
R55
R60
R58
Q3
R57
U6
C39
R54
C38
C37
C36
C41
C40
R62
R66
R67
C42
Q5
Q6
R61
C43
A1

Comment	Description	Designator	Footprint	LibRef	Quantity
Anten_IPEX		A1	Anten_IPEX	Anten_IPEX	1
BOOT	Nut Nhan 4 Chan 6x6mm	BT1	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
EN	Nut Nhan 4 Chan 6x6mm	BT2	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT3	Nut Nhan 4 Chan 6x6mm	BT3	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT1	Nut Nhan 4 Chan 6x6mm	BT4	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT2	Nut Nhan 4 Chan 6x6mm	BT5	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
BUT4	Nut Nhan 4 Chan 6x6mm	BT6	Nut Nhan 4 Chan Cam 6x6	Nut Nhan 4 Chan Cam 6x6	1
106 (10V/20%)	Dán	C1, C15, C19	tu 0805	Tu 0805 106 16V 10%	3
104 (10V/10%)	Dán	C2, C9, C10, C13, C14, C16, C27, C33	tu 0805	Tu 0805 104 50V 10%	8
104 0.1uF/275V	Dán	C3	tu 0805	Tu 0805 104 50V 10%	1
220uF	Dán	C4, C8	tu nhom 8 (e)	Tu Nhom 220uF 10V	2
106	Dán	C5	tu 0805	Tu 0805 106 16V 10%	1
105 (10V/10%)	Dán	C6, C22, C23, C24, C28, C29, C30	tu 0805	Tu 0805 105 50V 10%	7
100uF	Dán	C7	tu nhom 8 (f)	Tu Nhom 100uF 50V	1
22nF	Dán	C11	tu 1206	Tu 1206 22pF 50V 10%	1
6.8nF	Dán	C12, C18	tu 0805	Tu 0805 680pF 50V 10%	2
103/10V(10%)	Dán	C17	tu 1206	Tu 1206 104 50V 20%	1
470uF	Dán	C20, C26	tu nhom 10 (G)	Tu Nhom 470uF 35V	2
104 100nF	Dán	C21, C42	tu 0805	Tu 0805 104 50V 10%	2
33nF	Dán	C25, C41	tu 0805	Tu 0805 33nF 50V 20%	2
22pF	Dán	C31, C32, C45, C46, C47	tu 0805	Tu 0805 22pF 50V 10%	5
475 4.7uF	Dán	C34	tu 0805	Tu 0805 475 25V 20%	1
103/10V (10%)	Dán	C35	tu 0805	Tu 0805 104 50V 10%	1
104	Dán	C36, C48	tu 0805	Tu 0805 104 50V 10%	2
105	Dán	C37	tu 0805	Tu 0805 105 50V 10%	1
100uF	Dán	C38	tu 0805	Tu 0805 33nF 50V 20%	1
100uF	Dán	C39	tu nhom 6.3 (d)	Tu Nhom 100uF 25V	1
10pF	Dán	C40	tu 0805	Tu 0805 10pF 50V 10%	1
102 1nF	Dán	C43, C50	tu 0805	Tu 0805 102 50V 10%	2
471	Dán	C44	tu 0805	Tu 0805 471 50V 10%	1
33pF	Dán	C49	tu 0805	Tu 0805 33pF 50V 10%	1
Ethernet	Single Port RJ45 Connector With Magnetics Module And LED - Through Hole	CN1	RJ45-HR911105A-TH	RJ45-HR911105A-TH	1
XH1X9-P2.54-TH	Connector XH2.54 9P - Through Hole	CN2	XH1X9M-P2.54-TH	XH1X9-P2.54-TH	1
1N5819(SS14)-SM	1.0A Schottky Barrier Rectifiers - Surface Mount	D1	SMA-J_DO-214AC	1N5819(SS14)-SM	1
LESD5D5.0CT1G		D2, D3, D4	LESD5D5.0CT1G,	LESD5D5.0CT1G	3
1N5822(SS34)-SM	3.0A Schottky Barrier Rectifiers - Surface Mount	D5	SMA-J_DO-214AC	1N5822(SS34)-SM	1
1N5824(SS54)-SM	5.0A Schottky Barrier Rectifiers - Surface Mount	D6	SMA-J_DO-214AC	1N5824(SS54)-SM	1
ESP32-C5-WROOM-1/1U		ESP1	ESP32-C5-WROOM-1/1U	ESP32-C5-WROOM-1/1U	1
FUSE-CHIP-SM	Chip Fuses - Surface Mount	F1	FUSE-CHIP-1812-SM	FUSE-CHIP-SM	1
POWER-220V		HB1	HB9500	HB9500	1
JP esp	Cắm	JP1, JP2	JP 1x5 DT	JP 1x5 2.54 Duc Thang	2
JP esp	Cắm	JP3, JP4	JP 1x9 DT	JP 1x9 2.54 Duc Thang	2
GND/5V/SDA/SCL	Cắm	JP5	JP 1x4 DT	JP 1x4 2.54 Duc Thang	1
5V	Cắm	JP6	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
3V3	Cắm	JP7	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
GND	Cắm	JP8	JP 1x5 DT	JP 1x5 2.54 Duc Thang	1
JP 2x5 2.54	Cắm	JP9	JP 2x5 DT	JP 2x5 2.54 Duc Thang	1
10uH	EMI Filter Inductor - Through Hole	L1	L-UU9.8-L17.0-W12.0-H17.0-TH	L-FILTER-TH	1
33uH - 3A SMD	Dán	L2, L12	Cuon Cam 12x12x10	CUON CAM 100uH SMD	2
10uH	Chip Inductor - Surface Mount	L3	L-CHIP-0805-T1.05-SM	L-CHIP-SM	1
10D561-560V	Metal Oxide Varistor - Through Hole	MOV1	MOV-P7.5-D17.0-T3.6-TH	MOV-TH	1
15EDG - 4P - dt	15EDG - 4P - 3.81 mm đực thẳng	P1	15EDG - 4P - dt	15EDG - 4P - dt	1
Micro SD card TF5		pp1	MicroSD TF V5	Micro SD card TF5	1
S8050(J3Y)-SM	0.5A 25V NPN General Purpose Transistors - Surface Mount	Q1, Q2	SOT23-3_TO-236AB	S8050(J3Y)-SM	2
C1815	NPN transistor	Q3, Q4	SOT23	C1815 HF SMD	2
MMBT3904	NPN General Purpose Transistors 0.2A 40V - Surface Mount	Q5, Q6	SOT23-3_TO-236AB	2N3904(1AM)-SM	2
10k	Dán	R1, R14, R21, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R43, R44, R45, R46, R47, R49, R50, R52, R53	tro smd 0805	Tro 0805 5% 10k	27
5.1k	Dán	R2, R3	tro smd 0805	Tro 0805 1% 5.1k	2
1k	Dán	R4, R5, R25, R55, R56	tro smd 0805	Tro 0805 5% 1k, Tro 0805 1% 1k	5
1.21k	Dán	R6	tro smd 1206	Tro 1206 5% 1k	1
4.7k	Dán	R7, R8, R57, R58, R61, R62, R66, R67	tro smd 0805	Tro 0805 5% 4.7k	8
51R	Dán	R9, R11	tro smd 0805	Tro 0805 5% 30R	2
330R	Dán	R10, R12, R13, R16, R17, R18, R20, R22	Tro SMD 0805	Tro 0805 5% 330R	8
10R	Dán	R15	tro smd 0805	Tro 0805 5% 10R	1
62R	Dán	R19, R23	tro smd 0805	Tro 0805 5% 62R	2
22k	Dán	R24, R26	tro smd 0805	Tro 0805 5% 22k	2
7.5k	Dán	R42	tro smd 0805	Tro 0805 5% 4.7k	1
12k	Dán	R48	tro smd 0805	Tro 0805 5% 12k	1
1M	Dán	R51	tro smd 0805	Tro 0805 1% 1M	1
39R - Ferrite Beads	Dán	R54	Tro SMD 0805	Tro 0805 5% 390R	1
47k	Dán	R59, R60	tro smd 0805	Tro 0805 5% 47k	2
22R	Dán	R63, R64, R65	tro smd 0805	Tro 0805 5% 22R	3
2k	Dán	R68	tro smd 0805	Tro 0805 1% 2k	1
SIM SOCKET 6P		S1	SIM SOCKET 6P	SIM SOCKET 6P	1
10K	Cắm	T1	Bien Tro Vi Chinh Dung	Bien Tro Tinh Chinh 10K Dung	1
Type_C_16P		Type_C1	Type_C_16P	Type_C_16P	1
LM2976		U1	TO263	LM2596	1
cp2102		U2	cp2102	cp2102	1
74HC595-SM	8-Bit Shift Registers With 3-State Output Registers - Surface Mount	U3	TSSOP16-0.65-4.4MM	74HC595-SM	1
XL4005-SM	5A 180KHz 36V Buck DC to DC Converter - Surface Mount	U4	D2PAK-6_TO-263BA	XL4015-SM	1
W5500-SM	Ethernet ICs 3in1 Enet Controller TCP/IP +MAC+PHY - Surface Mount	U5	LQFP48-0.5-7X7MM	W5500-SM	1
A7677_openCPU	LTE CAT 1 module, 4G + (GNSS)	U6	A7677	A7677_openCPU	1
25Mhz	Crystal For all Crystal 3pin.	Y1	Thach Anh 2 chan cam	Thach Anh 2 chan Cam	1